# AI reinvention made real

## Accenture Innovation Challenge 2026

# Team details

| TEAM NAME: | |
|---|---|

## Name(Team Leader)

College:
Stream:
Year of graduation:

## Name

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Stream:
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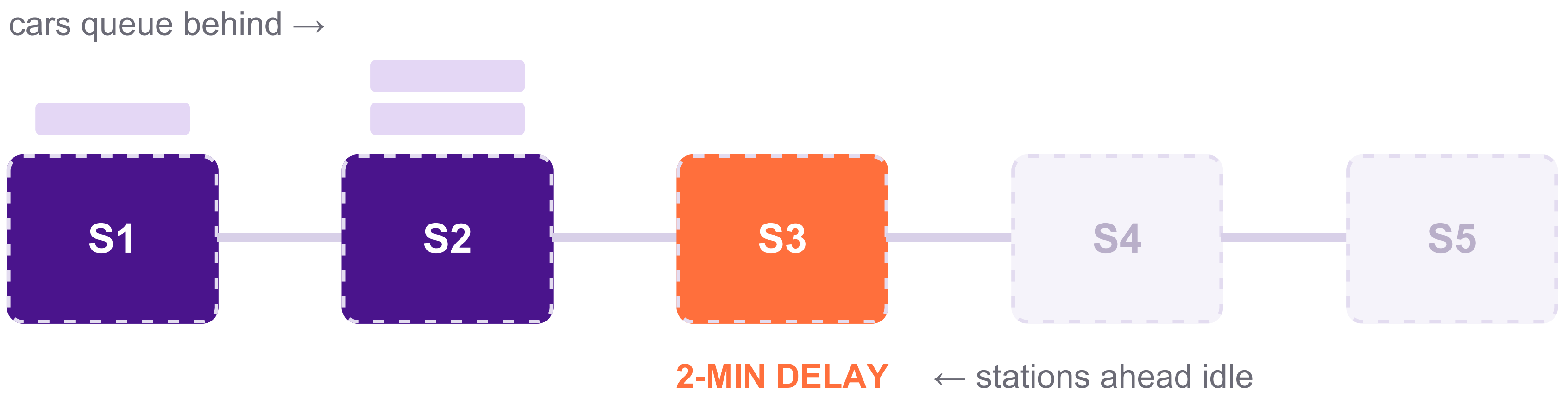
**Local Delay, Global Cost:** a slowdown at one station rarely stays local — it cascades into downstream cost, idle time, and undetected defects.

## THE RIPPLE EFFECT, STATION BY STATION




On a vehicle assembly line, work moves through dozens of stations at a fixed takt time. **When a station falls behind** a worn tool, a stuck fastener, a drifting process parameter vehicles queue up behind it while stations further down the line run out of work, and a fully stopped line can cost a plant tens of thousands of dollars a minute.

Bottlenecks aren't fixed to one station. Published research on automotive assembly systems shows they migrate depending on product mix and demand so static, shift end reporting can't reliably anticipate them.

Quality problems compound the same way. **The "1:10:100 rule"** a long standing quality principle  holds that a defect is roughly 10x costlier to fix at end of line than in process, and 100x costlier once it reaches the customer. Many stations, especially older ones, carry little digital instrumentation so the riskiest stations are often the least observed.

**1 : 10 : 100**

Relative cost to fix a defect in process vs. at end of line vs. after it reaches the customer

**30%+**

OEE improvement from real time bottleneck detection in a year long automotive powertrain pilot

**Dynamic**

Bottlenecks shift station to station with product mix static reports can't track them

Sources: Zhang, X. et al. (2023), "A digital twin-based method for production system performance prediction", ResearchGate • 1:10:100 rule, established quality-management principle (QCAdvisor, Invensis Learning)

**Modelling the Rhythm, Not the Geometry:** A lightweight Assembly Intelligence Layer that tracks flow and quality , no 3D factory replica required.

## THREE LAYERS, WORKING AS ONE SYSTEM

1

### The Line Mirror

A functional model not a 3D replica tracking cycle time vs. takt time per station, queue depth between stations, and each vehicle's process history.

2

### The Flow Brain

A statistical baseline plus continuous drift detection catches gradual creep before it's a failure, then forecasts how a deviation will affect downstream queues e.g., a station's buffer emptying in ~20 minutes.

3

### Defect Radar

Every vehicle carries a rolling risk score built from the process parameters it passed through, flagging it for a targeted check at the next gate not at the end of the line.

***Verified result:*** *a comparable real time bottleneck detection approach improved Overall Equipment Effectiveness by over 30% in a one year pilot at an automotive powertrain assembly line.*

## NO SENSORS? THREE WAYS WE INFER STATE ANYWAY

### Neighbor inference

Infer an unmonitored station's state from the flow rate of the stations flanking it

### Soft sensors

Power draw, a basic camera feed, or a single accelerometer stand in for a dedicated sensor

### Uncertainty modeling

Unmonitored stations are shown as a probability, not a false certainty  e.g., ‘68% likely normal, 32% running slow’

Sources: Lugaresi, G., Matta, A., Ragazzini, L. et al. (2024), “Digital Twin-based bottleneck prediction for improved production control”, Computers & Industrial Engineering • Zhang, X. et al. (2023), ResearchGate

# Thank you